

Sam Tran, Software Engineering

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PROFESSIONAL SUMMARY	Software Engineering graduate with experience developing AI-powered applications, RESTful APIs, and scalable full-stack systems. Strong interest in Artificial Intelligence, Machine Learning, and Cybersecurity, demonstrated through industry experience and an AI-assisted job recommendation platform using NLP techniques. Seeking graduate opportunities to apply software engineering and AI to real-world problems while continuing postgraduate study in Cybersecurity and Artificial Intelligence.	
EDUCATION	BEng, Bachelor of Engineering: Computer Science, Software Engineering Predicted Second-class honours, upper division (2:1 Hons) Master of Science Cybersecurity and Artificial Intelligence, University of Sheffield	Sheffield
EXPERIENCE	Jan 2025 — Mar 2025	Full-stack Collaboration Project, Elanco • Contributed to a full-stack web application using Flask (REST APIs) and React. • Developed Python automation scripts integrating AI models and APIs. • Deployed applications to Azure, supporting scalability and reliability. • Worked in an Agile team (sprint planning, stand-ups, retrospectives). • Wrote unit tests using unittest to reduce regressions.
	Mar 2023 — Aug 2023	Software Intern, Pro Check-Up • Built a Python-based search system to improve database query workflows. • Profiled and debugged performance bottlenecks using built-in tools. • Automated data extraction with Selenium. • Designed and optimized PostgreSQL schemas for efficient querying. • Provided technical support and troubleshooting for internal systems.
PROJECTS	AI-Assisted Job Matching Platform (Dissertation) • Developed a full-stack AI-powered job recommendation platform that analysed candidate skills and live job descriptions using Natural Language Processing (NLP). • Engineered a recommendation engine to rank vacancies based on semantic similarity rather than simple keyword matching. • Built RESTful APIs in Flask to integrate the AI recommendation engine with a responsive web application. • Implemented automated job ranking and personalised recommendations to improve match quality and explainability. • Designed the system architecture, relational database, API layer and evaluation methodology following software engineering best practices. • Applied Git-based version control, Agile development and systematic testing throughout the project.	
	Web-based Inventory Management System • Full-stack system using Laravel, PHP, JavaScript, and SQL. • Implemented backend models, controllers, and database schemas. • Built a responsive UI with Bootstrap. • Wrote PHPUnit tests to reduce post-deployment bugs.	
	Complaint Management System • Designed and implemented a layered front-end architecture using React and TypeScript, separating presentation (UI components), application/services, domain logic, and data access layers. • Enforced clear dependency boundaries using service modules and interfaces to reduce coupling and improve maintainability using Google Firebase • Managed state and side-effects through dedicated service layers to prevent business logic leakage into UI components. • Wrote unit tests for domain and service logic using Jest, improving testability independent of UI	
SKILLS	Python, JavaScript, TypeScript, C#, PHP, SQL, HTML, CSS Flask, React, Laravel, Node.js	PostgreSQL, MySQL, SQL Server Git, Azure, REST APIs, Unit Testing, Agile/Scrum